Hertie-Institut für klinische Hirnforschung

Neural Mechanisms of Voluntary Body Memory Control

DZNE Deutsches Zentrum für Neurodegenerative Erkrankungen in der Helmholtz-Gemeinschaft

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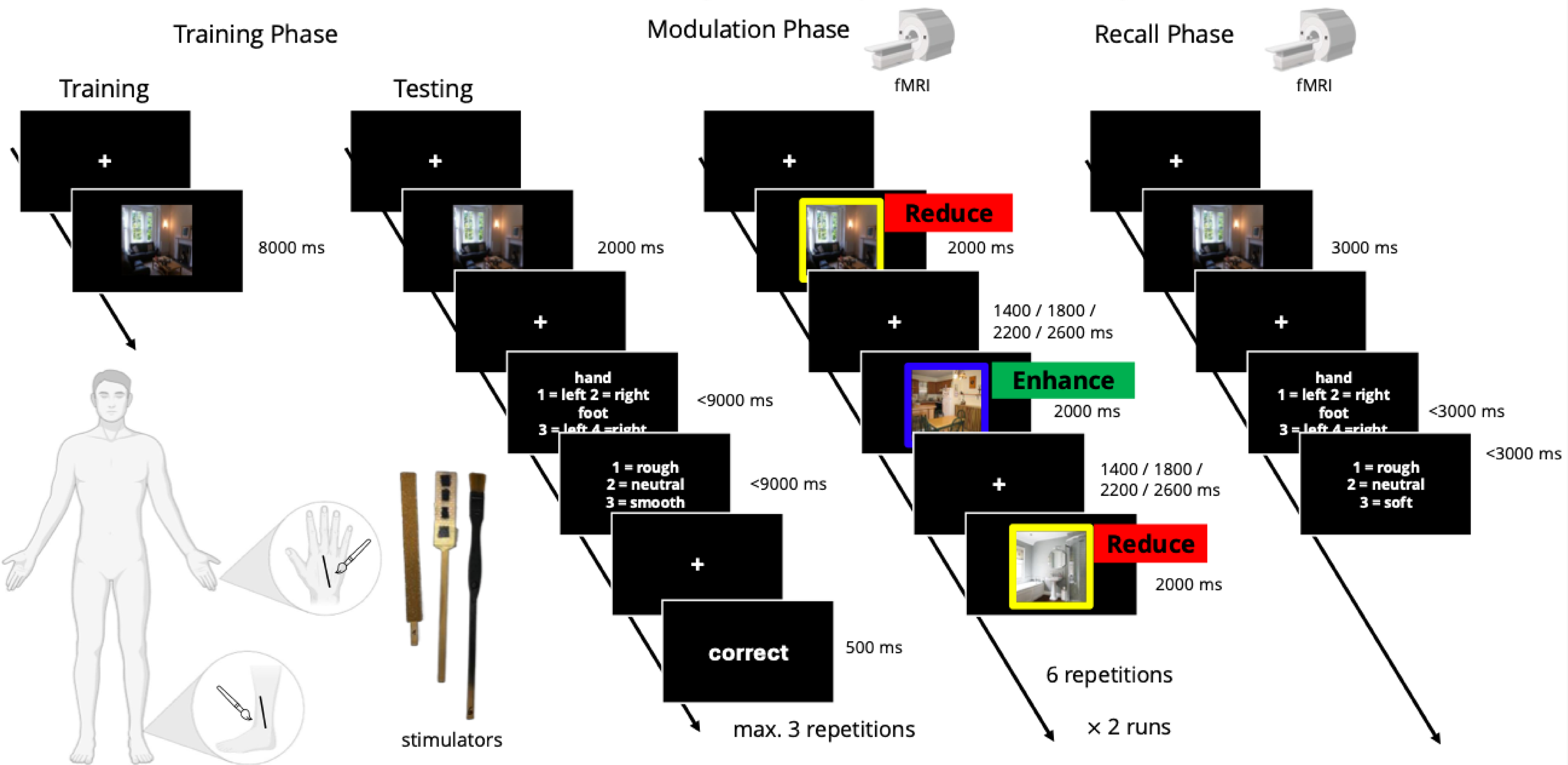
INTRODUCTION

1. Everyday bodily experiences form memory traces that influence our behaviour and mental health. The capacity to selectively retrieve or suppress memories represents a fundamental regulatory function of the human brain, supporting cognitive flexibility and emotional stability. [1]
2. Voluntary memory control is well characterised for verbal and visual memories through top-down prefrontal inhibition of hippocampal retrieval, but whether these mechanisms extend to body memories has been understudied. [2,3]
3. Body memories are inherently body-referenced and spatially specific, may imposing distinct demands on prefrontal-hippocampal circuits. [4-6]
4. Whether voluntary suppression of bodily memories produces lasting consequences for how those memories are subsequently reinstated is still unknown, which is important for understanding why intrusive bodily memories can resist conventional suppression in some conditions.

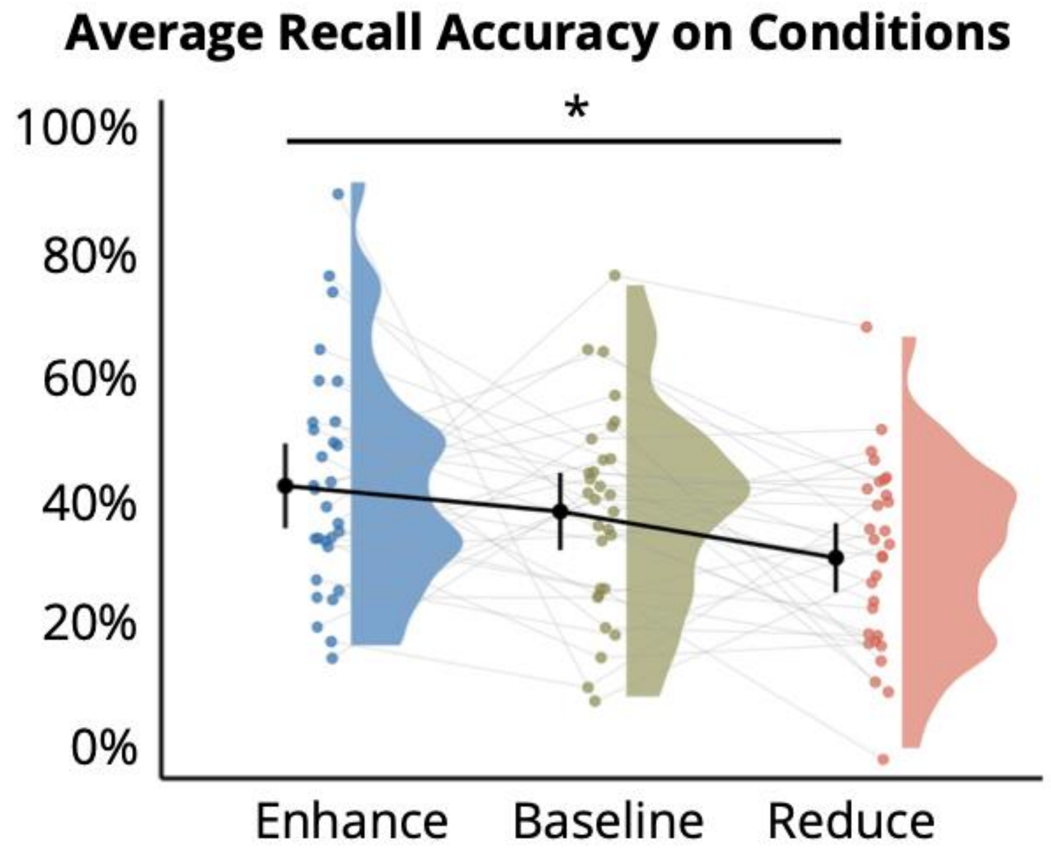
METHODS

- n = 30 (20 females)
- mean age = 24.03 years
- right-handed

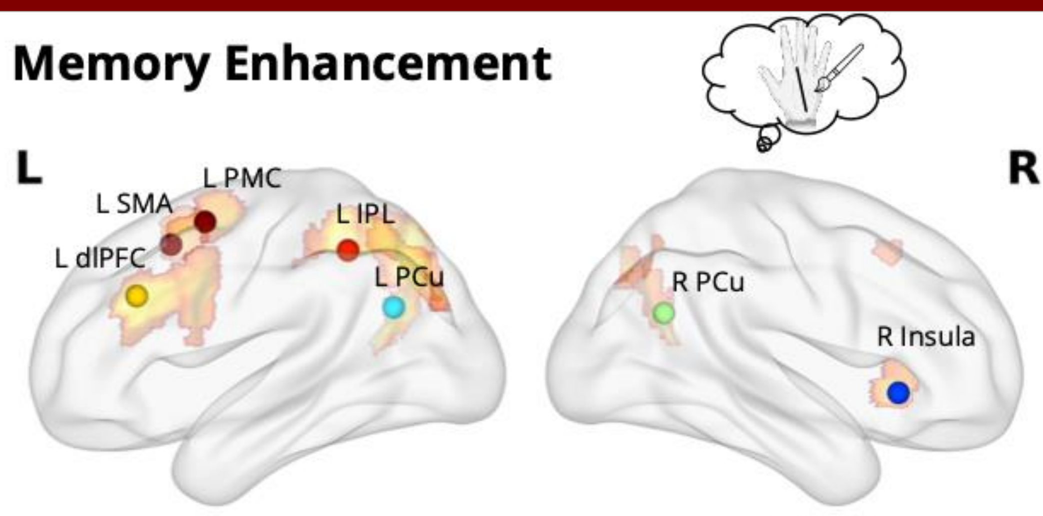


RESULTS

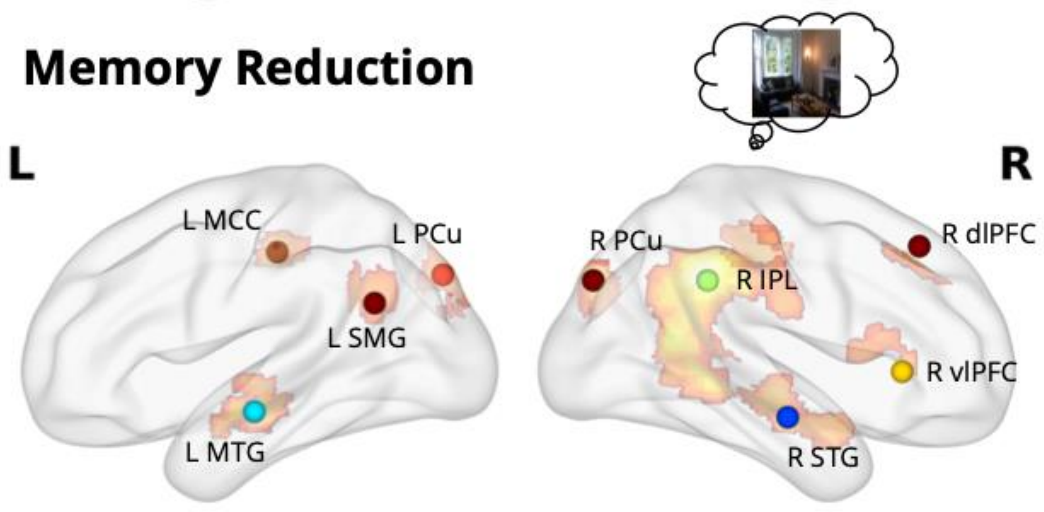
Average Recall Accuracy on Conditions



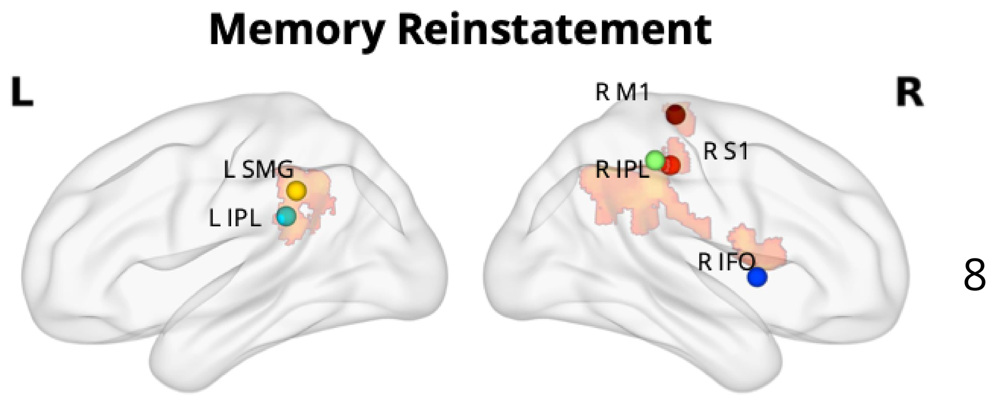
Memory Enhancement



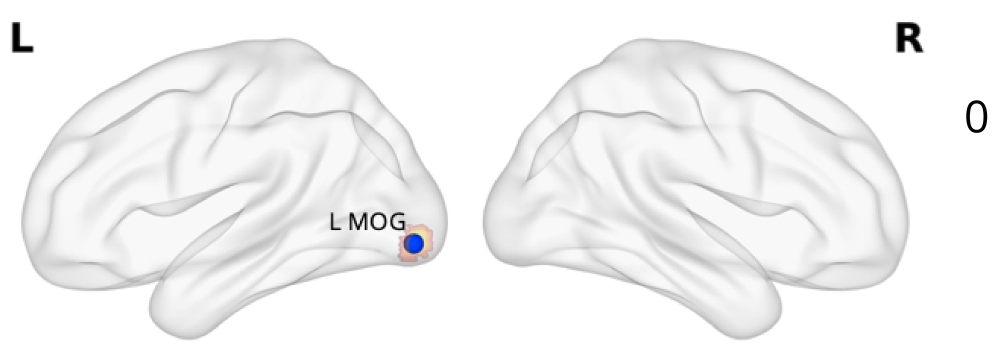
Memory Reduction



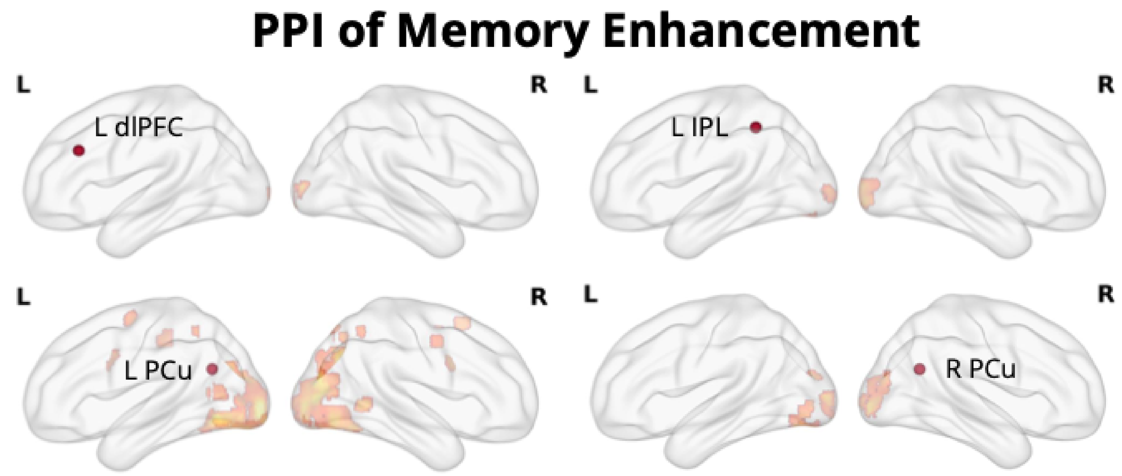
Memory Reinstatement



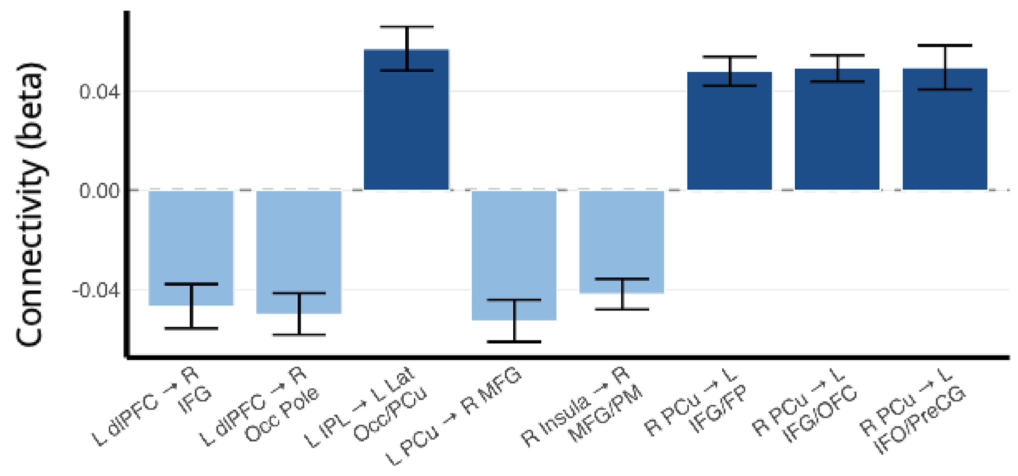
Individual Influence on Memory Reinstatement



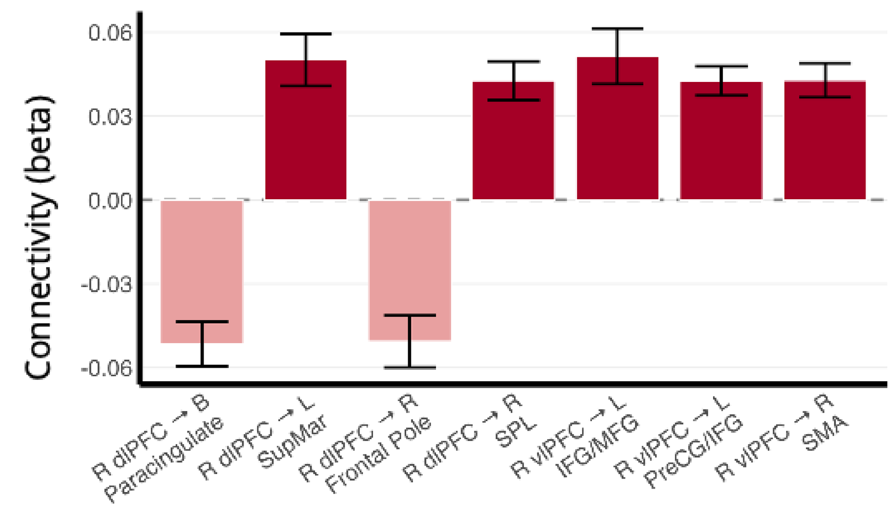
PPI of Memory Enhancement

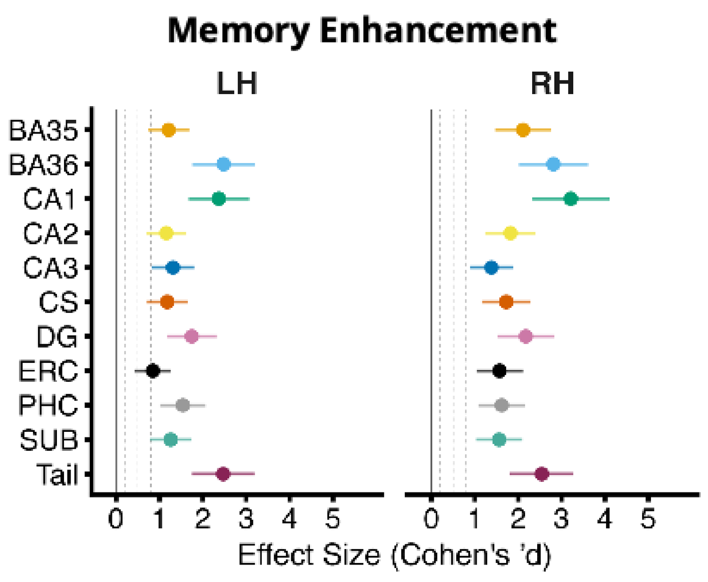


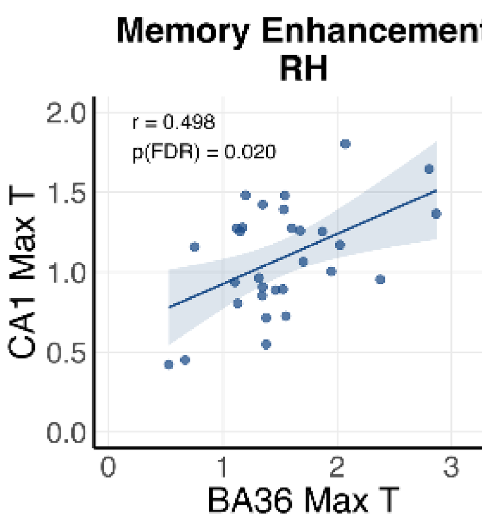
Functional Connectivity of Memory Enhancement

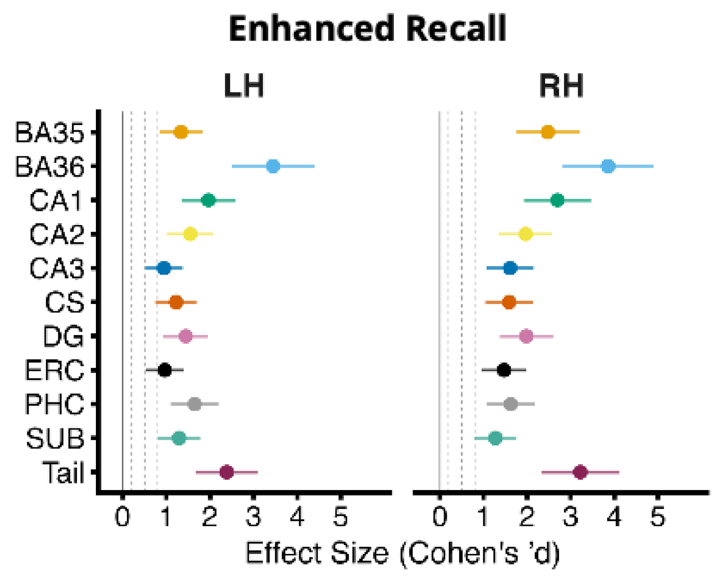


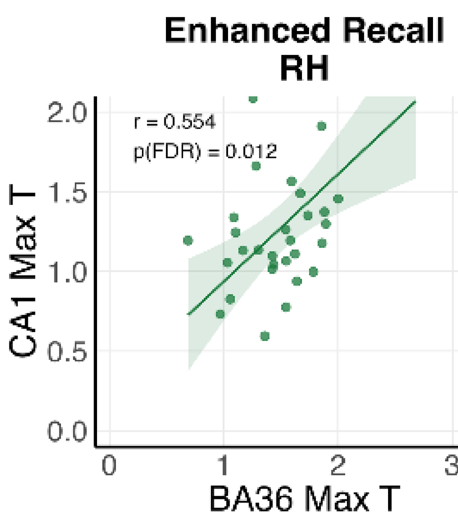
Functional Connectivity of Memory Reduction

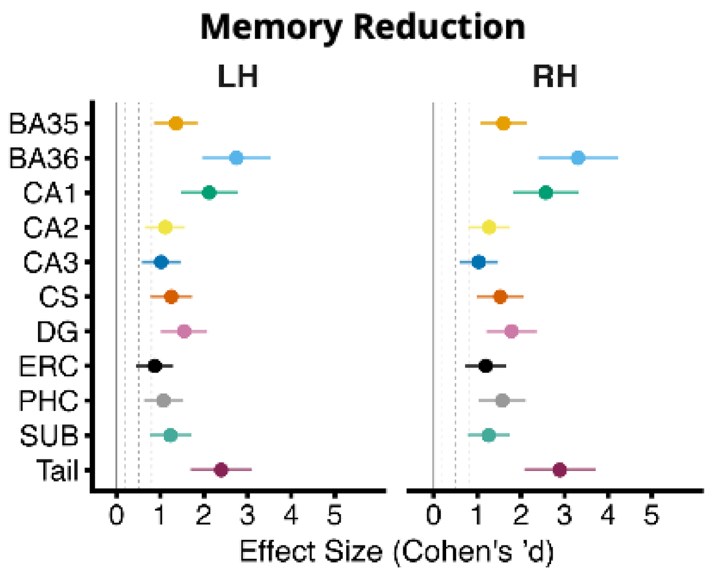


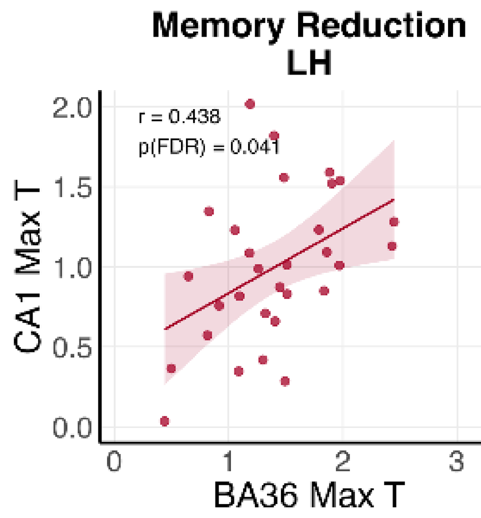


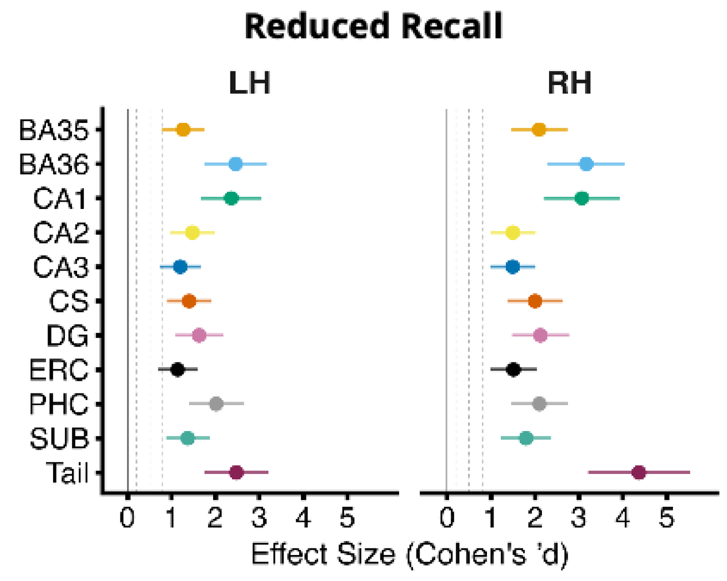


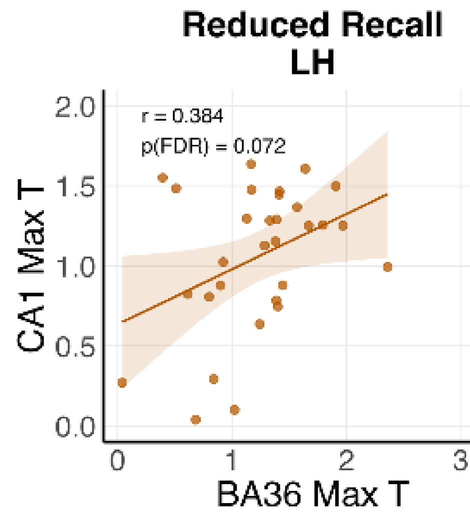












- **Body memories can Be voluntarily modulated and involves dissociable networks.**
- **Voluntary body memory control produces dissociable and lasting neural consequences at later recall.**
- **A MTL circuit is selectively recruited during voluntary body memory control.**

CONCLUSIONS

- Body memories can be voluntarily enhanced or reduced, establishing voluntary body memory control as a measurable human capacity.
- Memory enhancement recruits a left PCu-centred parietal-prefrontal network, while reduction recruits a right-dominant prefrontal network.
- Voluntary reduction renders body memories neurally indistinguishable from natural forgetting at later recall.
- A stable MTL circuit tracks body memory control, identifying potential therapeutic targets for intrusive body memories in related conditions such as somatic symptoms.

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